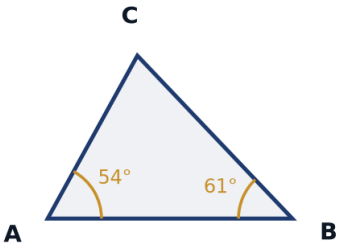
Angles of elevation and depression (Extended)

Core Concepts: Angles of elevation and depression (Extended)

1 In $\triangle ABC$ studying angles of elevation and depression (extended), $m\angle A = 37^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 98°

b. 108°

c. 90°

d. 143°

2 A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$?

a. 6

b. 11

c. 9

d. 12

Procedural Fluency & Algebraic Manipulation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 In $\triangle ABC$ studying angles of elevation and depression (extended), $m\angle A = 37^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 91°

b. 101°

c. 83°

d. 143°

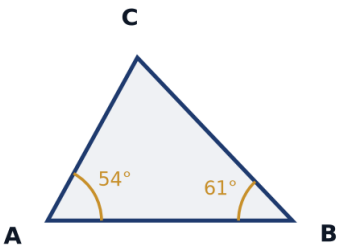
5 A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm

c. 24 cm

d. 54 cm



6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$?

a. 9

b. 14

c. 12

d. 15

Graphical & Geometric Interpretation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 In $\triangle ABC$ studying angles of elevation and depression (extended), $m\angle A = 37^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 84°

b. 94°

c. 76°

d. 143°

8 A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm

c. 30 cm

d. 96 cm

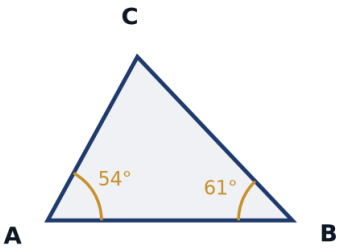
9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$?

a. 12

b. 17

c. 15

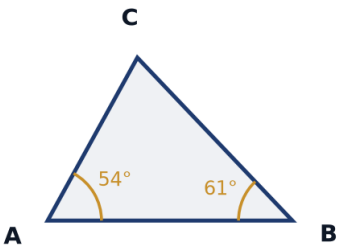
d. 18



Exam-Style Problem Solving & Modeling

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 In $\triangle ABC$ studying angles of elevation and depression (extended), $m\angle A = 37^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 77°

b. 87°

c. 69°

d. 143°

11 A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm

c. 36 cm

d. 150 cm

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$?

a. 15

b. 20

c. 18

d. 21

© ClipSAT: By Mr. Mohamed Abdallah +966597688647